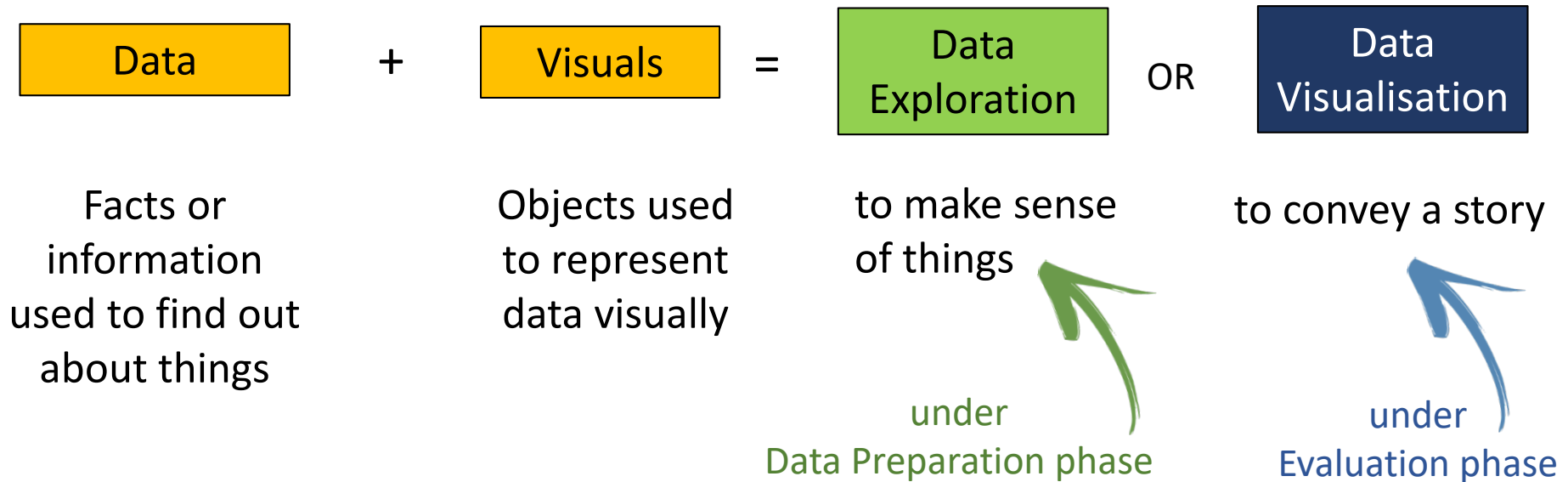




Data Understanding & Data Preparation: Exploratory Data Analysis (Use of Visuals to Assist)

What is Exploratory Data Analysis?

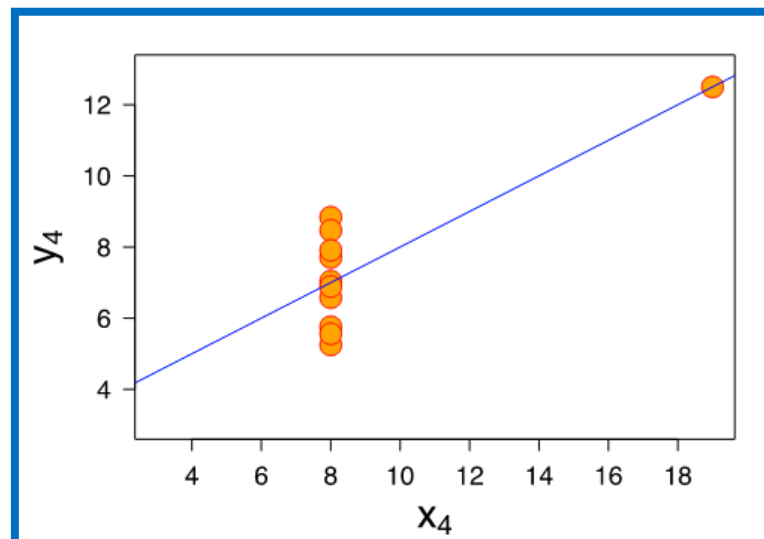
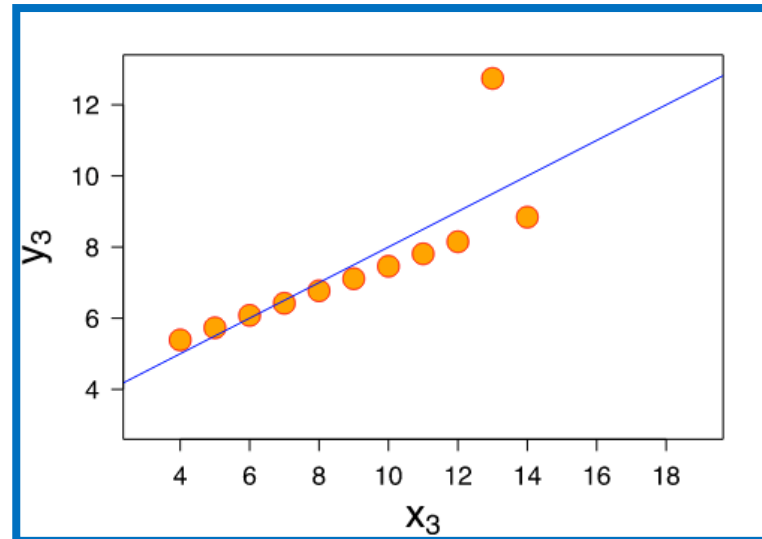
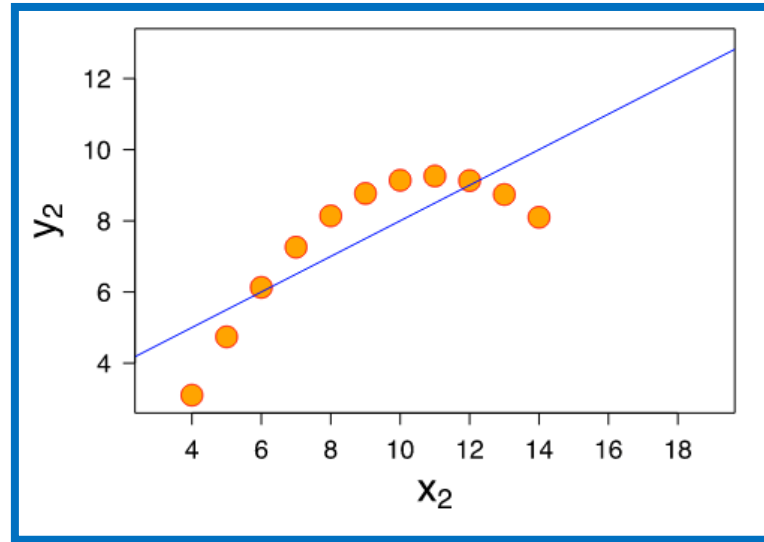
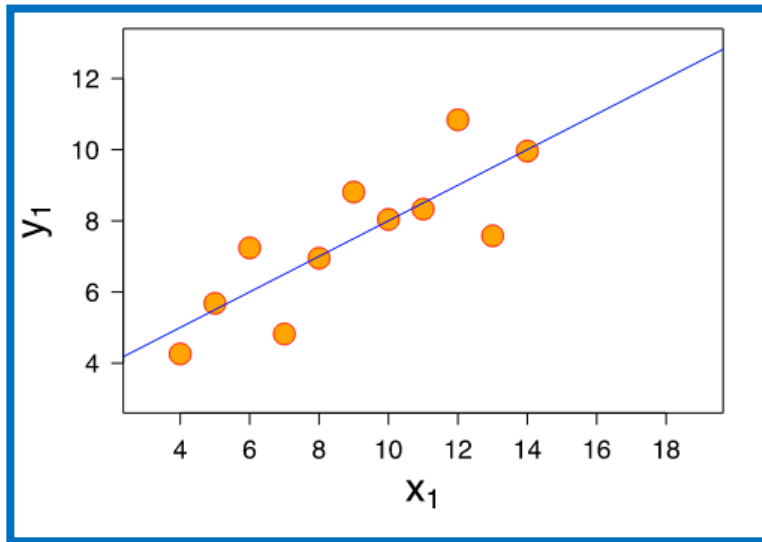


Anscombe's Quartet Data highlights the importance of visuals

Set I		Set II		Set III		Set IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

- The **mean** of $x = 9$ for each set
- The **mean** of $y = 7.5$ for each set
- The **variance** for $x = 11$ for each set
- The **variance** for $y = 4.12$ for each set
- The **correlation** between x and y is **0.816** for each set
- They are SAME yet DIFFERENT.

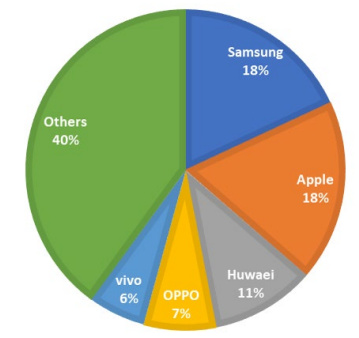
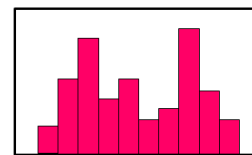
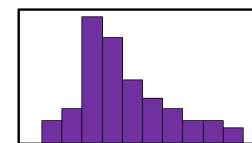
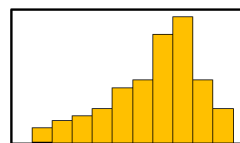
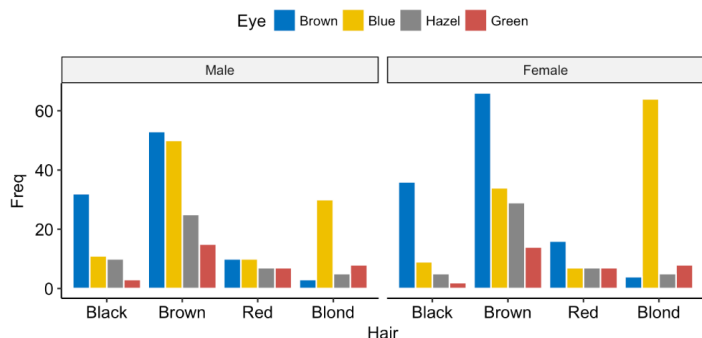
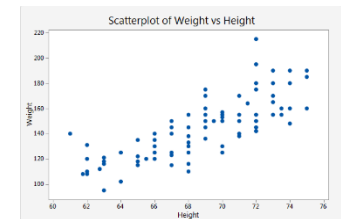
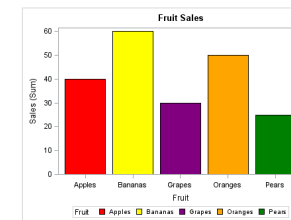
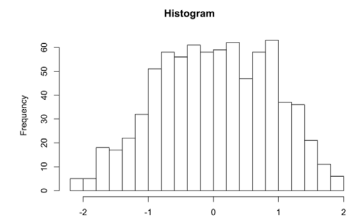
Visualizing the Data



Numerical and graphical representation of data

Perform easy and quick understanding about a data set through

- Univariate data analysis
- Bivariate analysis
- Multivariate analysis
- Understanding distributions and transformation
 - Normal probability distribution

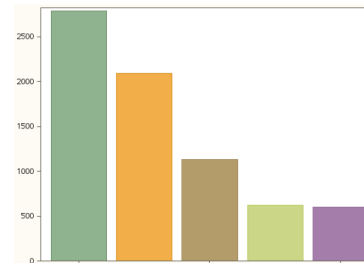
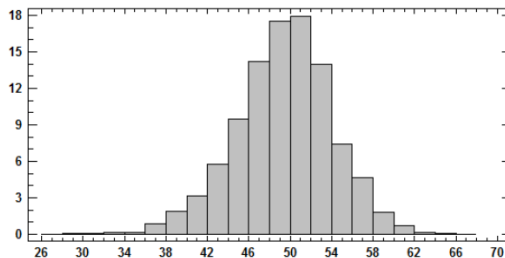
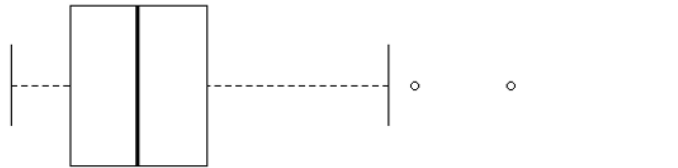


Univariate Analysis

- **Single-variable** or **univariate** data refers to data where we're only observing one aspect of something at a time.
- E.g. We take a group of people, measure their heights, and get this list of heights:
1.60m, 1.58m, 1.72m, 1.80m,
- This is univariate data, since we're only observing **one aspect** (the height) of each person.

Univariate Analysis

- For continuous/quantitative variables, numerical measures such as
 - averages (mean), variance, standard deviation, quantiles, median quartiles, and so on.
- Using these values to calculate e.g. skewness, kurtosis, spread, etc, to understand the distribution of the data, to find out if there is any biasness, outliers, and so on.



Names of these graphs?



Bivariate Analysis

- With **two-variable**, or **bivariate** data, we observe **two aspects**.
- We can put our observations into a table.
- E.g. We take a group of people, measure their heights and weights, and get the following information:

height	weight (lbs)
5'4"	140
5'7"	150
5'4"	130
6'	200

- This is bivariate data, since we have observations about two aspects (the height and weight) of **each person**.

Bivariate Analysis

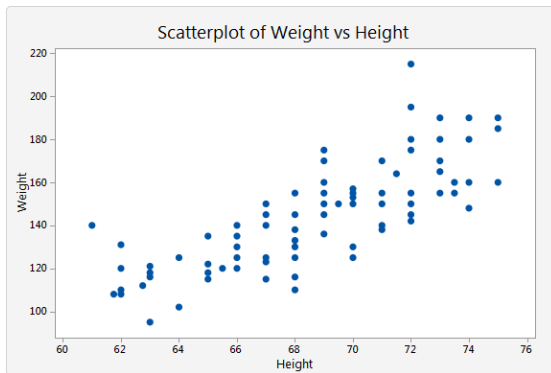
- Different variables require different graphs to represent the unique relationship.

Numeric-to-numeric
relationship

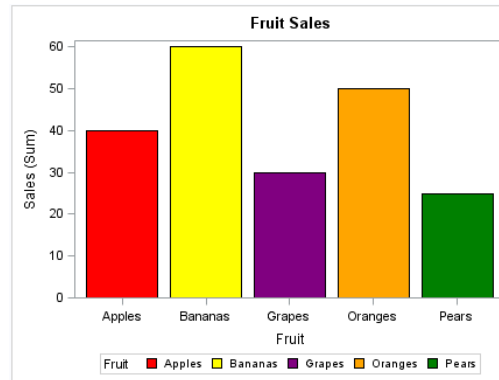
Numeric-to-categorical
relationship

Categorical-to-
categorical relationship

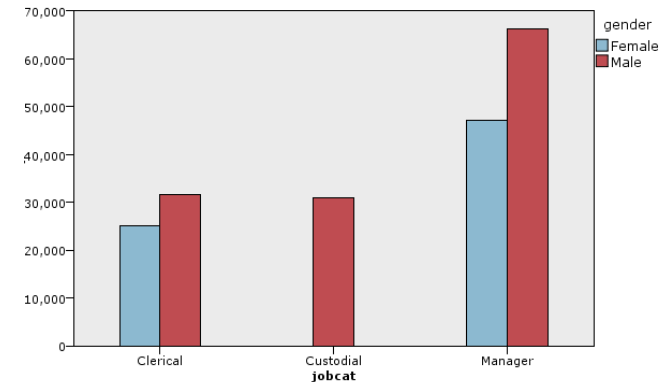
Scatterplot



Barplot

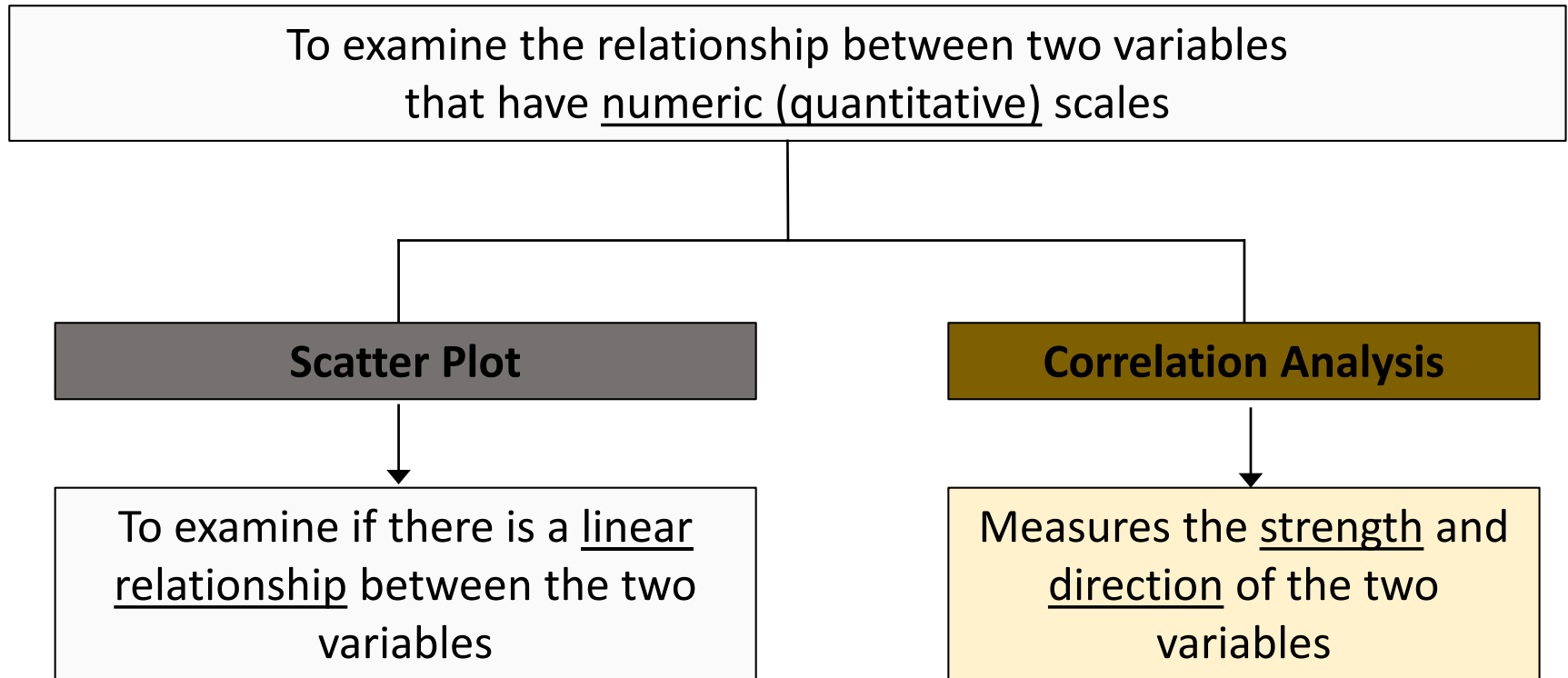


Frequency chart

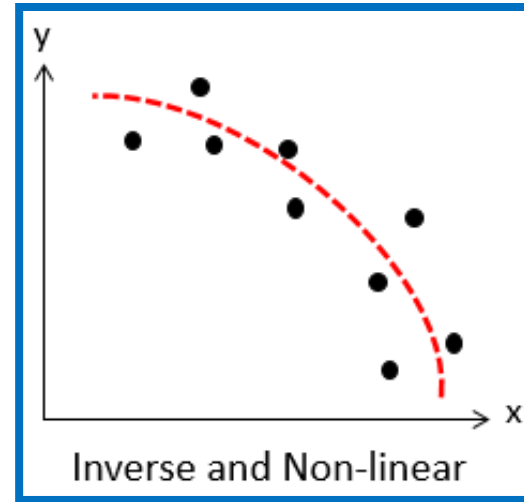
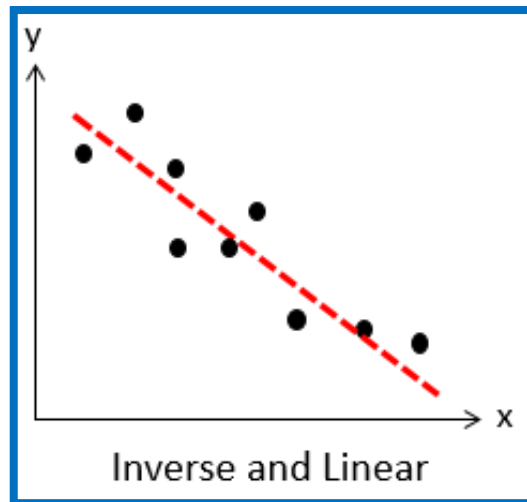
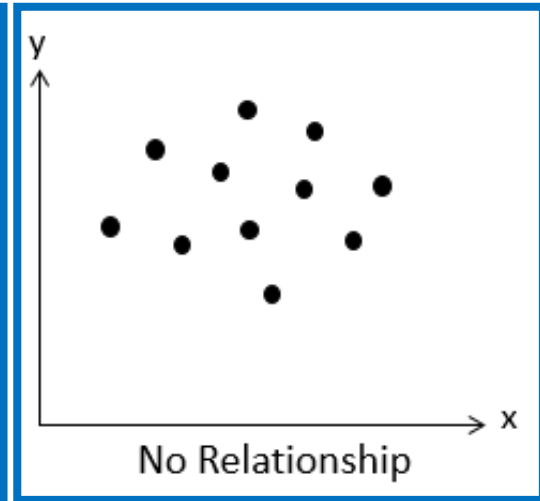
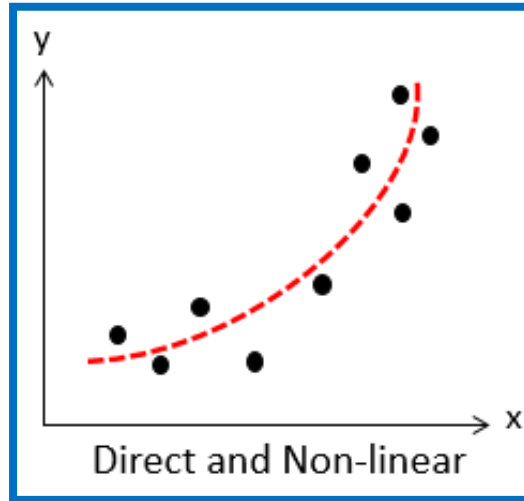
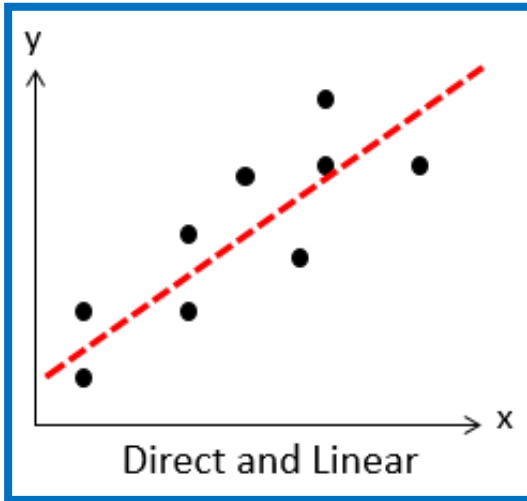


Bivariate Analysis

Apart from Scatter Plot, there is also the Correlation Heatmap



Start exploring existence of simplest linear relationship



There can be non-linear relationship, but since it is EDA, explore the simplest first



Properties of Pearson's correlation coefficient

Range The range of r varies from -1 to +1, through 0.

$$-1 \leq r \leq +1$$

Direction This refers to the **positive / negative sign** in front of r .

r is positive

y and x are **directly** related.

r is negative

y and x are **inversely** related.

Strength The **larger** the **absolute value of r** , the **stronger** the linear relationship between y and x .

$$|r| = 1$$

Perfect linear relation between y and x .

$$|r| = 0$$

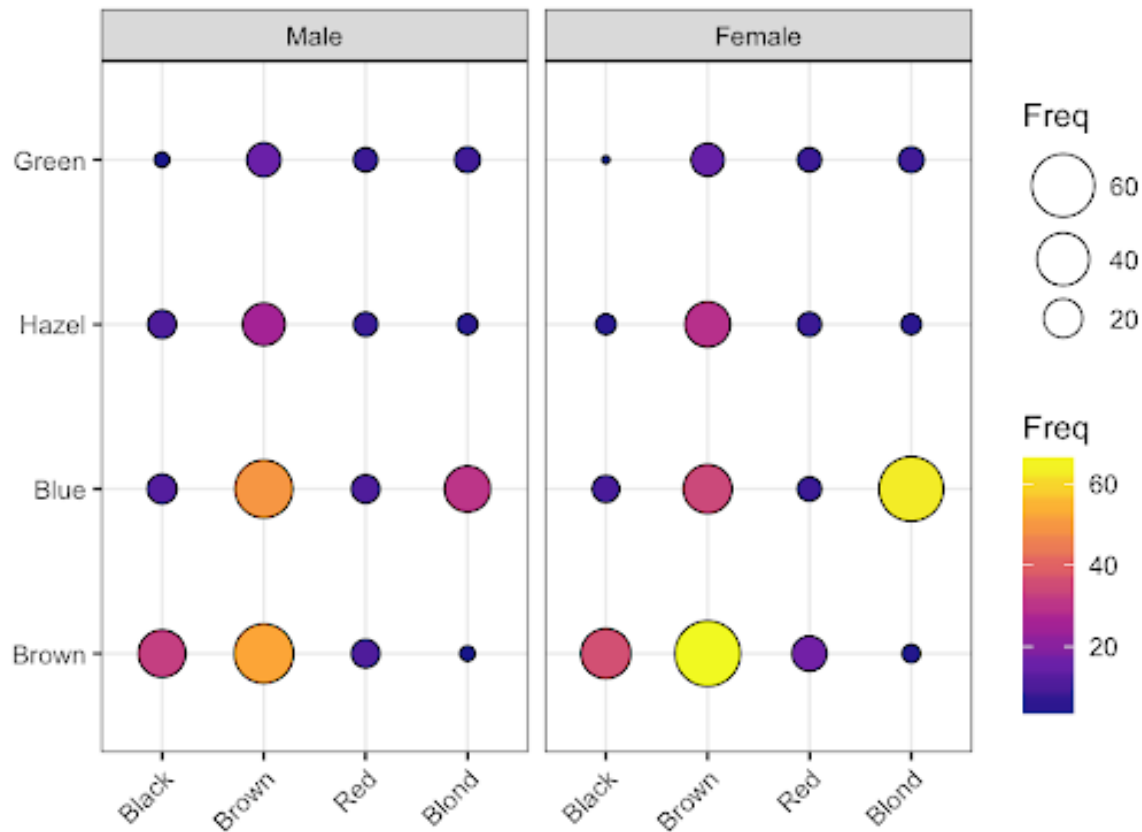
No linear relation between y and x .

$$0 < |r| < 1$$

Very weak $\rightarrow$ weak $\rightarrow$ moderately weak $\rightarrow$ moderately strong $\rightarrow$ strong $\rightarrow$ very strong

Multivariate Analysis

- The multivariate relationship is a statistical way of looking at **multiple dependent and independent variables** and their relationships.



Many different ways of representation.



The Need to know Data Distribution

- Parametric Vs Non-parametric

- Parametric data comes from a population that can be adequately modeled by a probability distribution (e.g. Normal/Gaussian, Binomial, Poisson) that has a fixed set of parameters.

❑ *(Interval or Ratio) Data typically follows a Normal Distribution*

❑ *Some models/algorithms makes assumptions of such distribution*

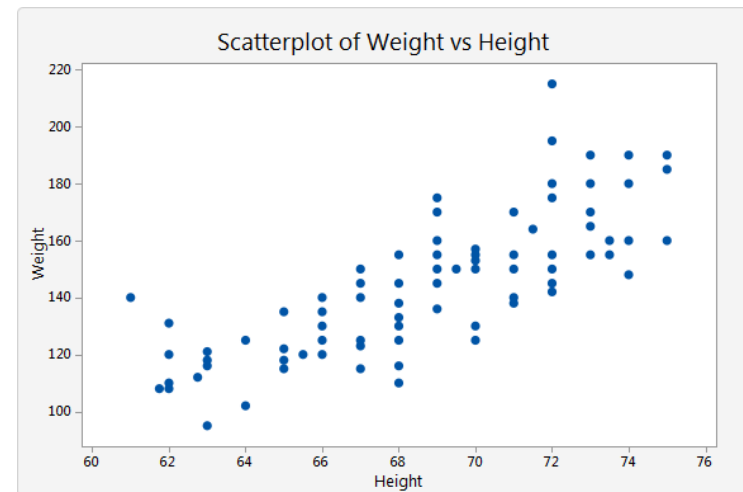
->linear regression

->hypothesis testing via t-test, ANOVA, etc

->Pearson's correlation coefficient, r .



Expect that the two attributes follow Normal Distribution before we can fit a straight line.



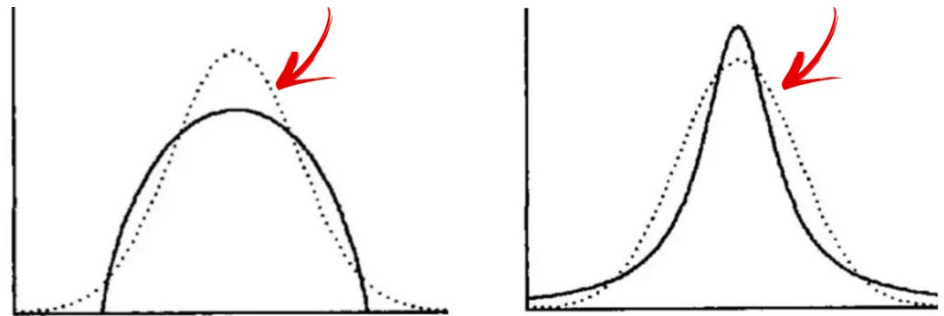
Test for Normality

- Specialised test

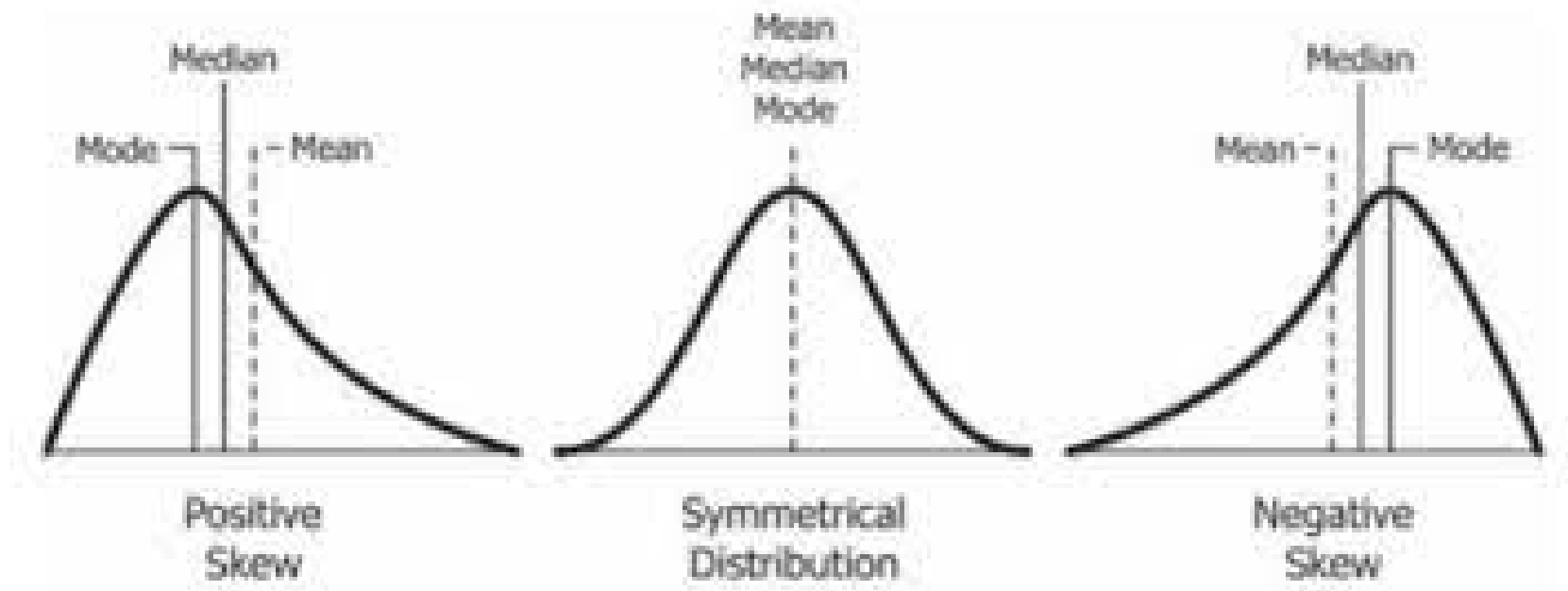
- Kolmogorov-Smirnov test
- Shapiro-Wilk test
- At least 10 ways in Python to do so

- Kurtosis and Skewness

- Visual inspection



What can skewness tell



**WE 
DATA
SCIENCE**